

CSLL: Complex Spectral Lead–Lag Networks for Delay-Aware Multivariate Time-Series Forecasting

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Abstract—Most multivariate forecasters that model dependencies between series do so in a phase-blind way: cross-variate attention, channel-mixing MLPs, and spectral or graph models represent inter-series coupling as real-valued and same-timestep, and cannot express that one series leads another by a delay. We ask a single question: when does modelling that lead–lag delay improve forecasts? We introduce a cross-channel operator applied per frequency band whose phase encodes a continuous inter-series delay (exact by the Fourier shift theorem), factorised to $\mathcal{O}(N)$ via additive delays and gated onto a channel-independent backbone. Two design choices make the mechanism testable rather than merely plausible: a zero-padded transform, so the phase is a linear (physical) shift rather than a circular one, and a direct synthesis of the shifted spectrum at the horizon, so the forecast-optimal delay equals the physical delay. On planted-delay data the operator recovers the true delays exactly (slope ≈ 1 , $r=1.00$) where a naive estimator collapses. Empirically we find a clear dichotomy: on data with delayed cross-series structure (weekly regional influenza) the phase branch lowers a lightweight model’s error by 13–19% over three seeds, closing most of the gap from a linear model to a cross-variate Transformer at an order of magnitude fewer parameters; on same-time-dominated data (5-minute traffic, standard long-horizon benchmarks) real-valued mixing suffices and the phase branch is redundant or mildly harmful. Because simple a-priori statistics of the inter-series lead do not cleanly predict the regime, we introduce a gated hybrid that runs a phase and a mixing branch and learns their blend: one model that, without a regime label, stays within seed noise of the best specialised variant everywhere and removes the same-time regression a phase-only model suffers. We report where the mechanism helps, where it is redundant, and where a strong backbone subsumes it, rather than a single headline number.

Index Terms—multivariate time series forecasting, cross-series dependence, frequency domain, lead–lag, spectral methods, deep learning

I. Introduction

Multivariate time-series forecasting (MTSF) predicts the joint future of N interacting series from shared history, underpinning traffic management, energy planning, and epidemiology. A central design question is whether—and how—to model dependencies across series. Channel-independent (CI) models such as PatchTST [1] and DLinear [2] forecast each series separately and are strikingly hard to beat [3]; channel-dependent (CD) models such as iTransformer [4] and Crossformer [5] mix across series for higher capacity. A recent line of critique [6, 7] shows that on the standard benchmarks these rankings are fragile and the datasets are largely self-predictable—which is exactly why CI models look so strong.

We focus on a specific, physically-motivated weakness shared by essentially all CD models: they represent cross-series coupling as real-valued and same-timestep. iTransformer attends over variate tokens at aligned times; channel-mixing MLPs mix instantaneously; spectral and graph models learn real-valued spectral graphs or a frequency-shared complex transform that encodes no phase. None can express that series j leads series i by a delay—even though propagation delays are physically ubiquitous (an upstream sensor leads a downstream one; an epidemic reaches regions in sequence). The methods that target lead–lag, LIFT [8] and VCformer [9], work in the time domain with discrete integer shifts; even the one frequency-domain cross-channel operator we are aware of, Fredformer [10], mixes channels per band with real-valued attention and encodes no delay.

Rather than assert that a delay-aware operator is better, we ask when it is. Our contribution is a mechanism that makes the question answerable, an honest map of its answer, and a model that acts on that map.

a) Contributions:

- 1) A complex spectral lead–lag operator (Sec. III, Fig. 1) with an exact-delay property, made physical and identifiable by a zero-padded linear-shift transform and a head-free direct forecast, so the forecast-optimal delay is the physical delay.
- 2) A correlation-initialised estimator that escapes the oscillatory time-delay-estimation loss surface and recovers planted delays exactly, where a naive gradient-only estimator collapses (Sec. IV, Fig. 2).
- 3) An empirical characterisation of when phase-aware coupling helps—a clear dichotomy between delayed-structure data (epidemics) and same-time data (traffic, standard benchmarks)—and the honest negative result that simple a-priori lead statistics do not predict the split (Sec. V).
- 4) A gated hybrid (Eq. 2) that adapts across the dichotomy: without a regime label it stays within seed noise of the best specialised variant in each regime, keeping the epidemic gain while removing the same-time regression a phase-only model suffers (Sec. VII). This, with its efficiency, is the practical model contribution.

II. Related Work

a) Classical and deep forecasters: Classical multivariate forecasting rests on vector autoregression and the

Box–Jenkins tradition [11, 12]; VAR remains a useful linear cross-series baseline. Deep long-horizon forecasting was reshaped by efficient Transformers—Informer [13], Autoformer [14], FEDformer [15], Pyraformer [16]—and by decomposition and basis-expansion models such as N-BEATS [17], N-HiTS [18] and TimesNet [19]. Autoformer’s Auto-Correlation aggregates within-series periods, not inter-series lead–lag—a distinction central to our work.

b) Channel strategies: DLinear/NLinear [2] and PatchTST [1] are channel-independent; iTransformer [4], Crossformer [5] and TSMixer [20] are channel-dependent, with TiDE [21] and RLinear [22] as strong MLP/linear points on the spectrum. Han et al. [3] frame the CI/CD trade-off as capacity versus robustness under distribution shift, and show it is not fundamental. Lightweight models push cost down—SparseTSF [23], CycleNet [24], SOFTS [25]—and channel clustering [26] interpolates CI and CD. CSLL’s gated additive branch is a spectral channel-partiality mechanism, and the hybrid makes the $CI \leftrightarrow CD$ position itself learnable per dataset.

c) Frequency-domain and cross-variate models: A large body of work forecasts in the spectral domain. FEDformer [15] and FiLM [27] operate on selected frequency modes; FITS [28] is a tiny complex-valued interpolator; FilterNet [29] and FilterTS [30] learn frequency filters; FreTS [31] applies complex MLPs with weights shared across all bins and hence carries no phase. Spatio-temporal graph models—Graph WaveNet [32], MTGNN [33], StemGNN [34]—and FourierGNN [35] learn real-valued (spectral) graphs, and Fourier neural operators [36, 37] provide the general operator-learning backdrop. The closest to us, Fredformer [10] and FreEformer [38], mix channels inside the spectrum but with real-valued or data-driven attention, encoding no explicit, interpretable delay; SimpleTM [39] uses phase-sensitive wavelet tokens but not as an inter-series delay. CSLL differs by making the phase of the cross-channel operator an explicit, bounded, continuous delay with an additive $\mathcal{O}(N)$ factorisation.

d) Lead–lag and cross-variate delay: A small but growing line targets inter-series lead–lag directly. LIFT [8] detects leading indicators by cross-correlation, estimates an integer leading step per channel at each time, and shifts the lagged variates to align them—a time-domain add-on over a backbone. VCformer [9] integrates multi-step lagged cross-correlations inside the attention weights, again in the time domain, and DeformTime [40] learns dependencies across variables at different time steps through deformable-attention offsets. All three are discrete, time-domain, and—by working through learned shifts or attention—do not expose a continuous, bounded delay that can be read off and checked against physics. Our operator instead places the lead–lag entirely in the phase of a per-band complex coupling: a single real number per channel pair, continuous (including fractional delays), bounded to the identifiable range, and factorised additively so the cost stays $\mathcal{O}(N)$. To our knowledge this specific combination—phase-as-continuous-delay with an additive $\mathcal{O}(N)$ factorisation and a head-free direct

forecast—is not present in prior work.

e) Spatio-temporal and epidemic forecasting: Our evaluation uses datasets with known geography: METR-LA/PEMS-BAY [41] and PEMS04/08 [42] carry road distances; the Cola-GNN regional influenza sets [43] carry adjacency graphs. We use these graphs only to validate learned delays, not to train, so the comparison to graph models [44] concerns interpretability, not leaderboard position.

f) Foundation models and evaluation: Pretrained forecasters [45–48] are largely univariate-per-channel or use generic attention for cross-series structure. Evaluation critiques [6, 7] motivate our protocol: a uniform training budget, reported significance, and datasets chosen for genuine cross-series structure.

III. Method: CSLL and the Gated Hybrid

a) Intuition: Shifting a signal in time does not change how much of each frequency it contains, only when each frequency’s oscillation peaks. That timing offset is a phase, and it grows linearly with frequency at a rate equal to the delay—so one number per pair of series captures the delay at every frequency at once. Modelling cross-series coupling as a per-band complex weight whose magnitude is the coupling strength and whose phase is a frequency-linear ramp therefore expresses “series i is a scaled, delayed copy of series j ” exactly and cheaply. The rest of this section makes that precise and addresses the two ways a naive implementation breaks it (a circular rather than linear shift, and a projection head that hides the delay).

b) Setup: Given a look-back $X \in \mathbb{R}^{N \times L}$ we predict $\hat{Y} \in \mathbb{R}^{N \times H}$. We use a zero-padded real DFT of length $M = 2L$, giving $F = M/2 + 1$ bins at $\omega_f = 2\pi f/M$; the padding makes the phase ramp a linear shift.

c) Spectral lead–lag branch: With $\tilde{X} = \text{rFFT}(\tilde{X})$, for band b and bin $f \in \mathcal{B}_b$,

$$W_b(f)_{ij} = (A_b)_{ij} e^{-i\omega_f(D_b)_{ij}}, \quad (D_b)_{ij} = p_i - q_j, \quad (1)$$

with $p_i, q_j = \frac{\tau_{\max}}{2} \tanh(\cdot)$ so $|(D_b)_{ij}| \leq \tau_{\max}$ stays inside the wrap-free identifiable range. The additive form factorises (1) into a per-channel source phase, a coupling matmul (A_b full, or low-rank $U_b V_b^\top$ for large N), and a per-channel target phase— $\mathcal{O}(N)$ phase work. A small controller can make p_i, q_i input-dependent (the dynamic variant).

d) Direct forecast: Rather than reconstruct the look-back and map it to the horizon with a head, we synthesise the shifted spectrum directly at $t \in [L, L + H)$. With a head, the loss-optimal delay for a pair with physical lag τ is $\tau - h$ (the head absorbs the horizon); the head-free direct forecast makes the forecast-optimal delay equal τ . Positions beyond the representable lead ($h \geq \tau$) read the zero padding, and the gated backbone covers that residual.

e) Exact-delay property and identifiability: For a zero-padded signal the linear shift by τ has DFT $e^{-i\omega_f \tau} \hat{S}(f)$ exactly (no wraparound), so if within a band series i is a scaled delayed copy of j , one (a, τ) pair

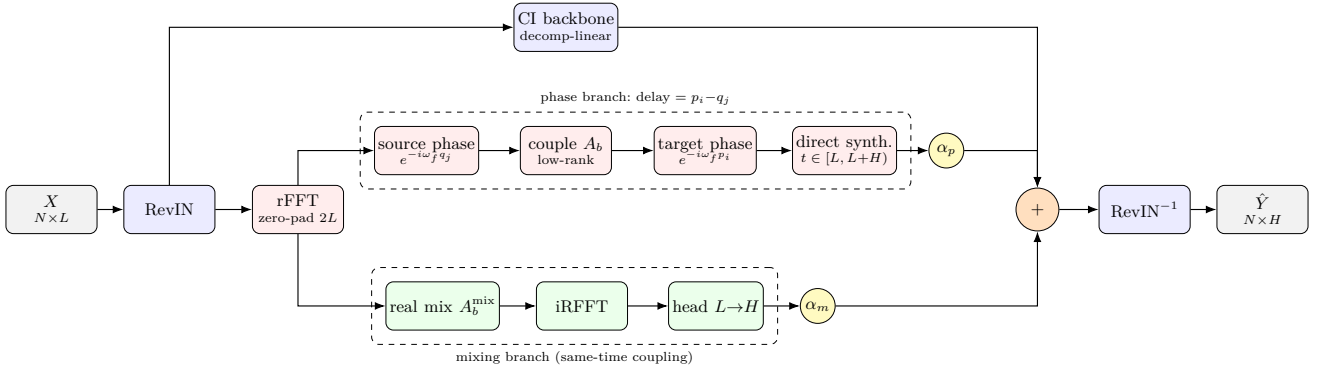


Fig. 1. CSLL hybrid architecture. A channel-independent backbone is corrected by two gated spectral branches. The phase branch applies a per-band complex operator whose phase encodes a continuous inter-series delay $p_i - q_j$ and forecasts by synthesising the shifted spectrum directly at the future positions; a zero-delay band reads the zero padding and contributes nothing, so it cannot express same-time coupling. The mixing branch supplies that via real-valued spectral coupling and an $L \rightarrow H$ head. Warm gates α_p, α_m let the data choose the blend, so one model spans delayed-structure and same-time regimes.

reproduces its contribution at every frequency. A real mixer ($D_b \equiv 0$) can only represent $\tau = 0$. Sharing (A_b, D_b) across a band pins $|A_b|$ (flat magnitude) and D_b (phase slope). The additive gauge $(p, q) \rightarrow (p + c, q + c)$ leaves D unchanged; we fix it by mean-centring the delay read-out.

f) Estimator: The MSE is oscillatory in D (period set by the dominant signal period), so gradient descent from $D = 0$ finds the nearest local minimum—the classic time-delay-estimation trap. We initialise p, q from lagged cross-correlation (a two-pass global estimate, or a weighted least-squares solve on the pairwise-lag graph for network data), then refine by gradient training. Sec. IV shows this is necessary and sufficient for exact recovery.

g) Hybrid: one model across the dichotomy: Because the phase branch cannot express same-time coupling, forcing all correction through it is harmful on same-time data. We run both a direct-phase branch and a real mixing branch (real spectral coupling, full-window synthesis, then an $L \rightarrow H$ head), each behind its own warm gate (Fig. 1):

$$\hat{Y} = \hat{Y}^{\text{base}} + \alpha_p \hat{Y}^{\text{phase}} + \alpha_m \hat{Y}^{\text{mix}}, \text{ then } \text{RevIN}^{-1}. \quad (2)$$

Where the phase branch does not lower validation error its gate decays toward zero and mixing carries the correction; where it does, the phase gate grows. The mixing branch adds only a low-rank coupling and one head ($\mathcal{O}(Nr + LH)$ parameters).

h) Complexity: The per-band operator costs $\mathcal{O}(N^2 F)$ dense, or $\mathcal{O}(NrF)$ with the rank- r coupling; the additive delays contribute $\mathcal{O}(N)$ phase work per bin rather than a per-bin $N \times N$ operator. With the real-arithmetic (cos/sin) DFT the whole model is real matmuls and is device-agnostic. The operator adds $B(2Nr + 2N)$ parameters—a few tens of thousands for the datasets here—against the hundreds of thousands of a cross-variate Transformer, and trains several times faster (Sec. VII). Delays are bounded by $\tau_{\max} = L/2$, so the representable lead scales with the look-back.

Algorithm 1 CSLL hybrid forward pass

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1:  $\tilde{X} \leftarrow \text{RevIN}(X)$ ;  $\hat{Y}^{\text{base}} \leftarrow \text{DecompLinear}(\tilde{X})$ 
2:  $(\tilde{X}_r, \tilde{X}_i) \leftarrow \text{rFFT}_{2L}(\tilde{X})$ 
3: for band  $b$ , bins  $f \in \mathcal{B}_b$  do ▷ phase branch
4:   apply source phase  $e^{-i\omega_f q}$ , couple  $A_b$ , target phase  $e^{-i\omega_f p}$ 
5: end for
6:  $\hat{Y}^{\text{phase}} \leftarrow \text{Re iDFT}_{t \in [L, L+H]}(\tilde{Z})$  ▷ direct
7:  $\hat{Y}^{\text{mix}} \leftarrow \text{head}(\text{iRFFT}(A^{\text{mix}} \tilde{X}))$ 
8: return  $\text{RevIN}^{-1}(\hat{Y}^{\text{base}} + \alpha_p \hat{Y}^{\text{phase}} + \alpha_m \hat{Y}^{\text{mix}})$ 

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TABLE I
Kill-test (leader–follower, planted delays): the v2 estimator recovers delays exactly; ablating any fix breaks it.

Estimator	test MSE	rec. slope	r
naive (cold gate, circular)	0.527	−0.02	−0.18
+ fixes, no corr. init	0.879	0.02	0.36
full estimator	0.154	0.99	1.00
information floor	≈ 0.147	—	—

IV. Does the Estimator Recover Delays?

We construct a leader–follower regime: a white-noise leader drives followers that are pure delayed copies with planted lags τ_i , so each follower is forecastable only through the correctly delayed cross-series link. Table I and Fig. 2 report test MSE and delay recovery. The naive estimator (no correlation init, circular shift, cold gate) fails—MSE far above the information floor, recovery slope ≈ 0 . The full estimator reaches the floor with exact recovery (slope ≈ 1 , $r = 1.00$), and each fix is individually necessary (removing the correlation init alone reverts recovery to ≈ 0). This isolates estimation from the downstream question of whether real data has recoverable delays.

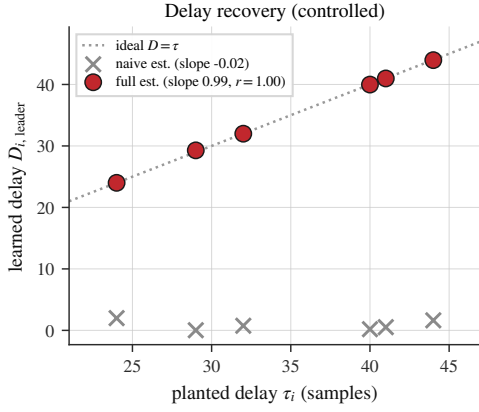


Fig. 2. Controlled delay recovery: the full estimator (red) lands on the identity; the naive estimator (grey) collapses to near-zero.

V. When Does Phase Help? An Empirical Dichotomy

a) The dichotomy: We measure the phase branch’s value directly as the relative test-MSE change from a real-valued mixer to the phase operator, holding everything else fixed (Table II). The split is clean by data, not model: on all three regional influenza datasets the phase branch helps substantially (+13–19%), while on 5-minute traffic and the standard benchmarks it is neutral-to-harmful (up to a 17% regression on Weather). Influenza carries delayed cross-region structure—an epidemic reaches regions in sequence over weeks—that a same-time mixer cannot represent but a complex, phase-carrying operator can; traffic and the benchmark series are dominated by same-time coupling, where the extra phase capacity only adds variance.

b) An honest negative: lead statistics do not predict the split: It is tempting to seek a cheap a-priori statistic that says which regime a dataset is in. We tried two. A lead/horizon ratio $\bar{\tau}/H$ (mean cross-correlation lag over coupled pairs) over-predicts a phase benefit on traffic. A delay advantage $\Delta(H) = \mathbb{E}_i[\rho_{\geq H}(i) - \rho_0(i)]$ —whether a partner leading by at least H predicts series i better than any same-time partner—correctly marks traffic as a mixing regime ($\Delta \ll 0$) but also comes out negative on most influenza settings, where the phase nonetheless helps (the benefit there is the added expressiveness of complex over real mixing, not a single dominant leading partner). Across our settings $\text{sign } \Delta(H)$ agrees with the outcome in fewer than half of settings (Table II, Fig. 3). We therefore do not offer a predictive scalar; we take the practical route (Sec. VII): a hybrid that learns the blend per dataset, removing the need to predict the regime.

VI. Experimental Setup

a) Datasets: We evaluate on twelve public datasets in three families (Table III). Two carry ground-truth geography that we use to validate learned delays: 5-minute freeway traffic (METR-LA, PEMS-BAY [41], with sensor road-distance graphs) and weekly regional influenza (US-state, Japan-prefecture, US-region, with adjacency graphs [43]). The standard long-term forecasting (LTSF)

TABLE II
When phase beats mixing: measured phase-vs-mixing gain per setting. The split is clean by data, but the a-priori statistic $\Delta(H)$ predicts it in fewer than half of settings (see footnote).

Dataset	H	phase gain (%)	regime	$\Delta(H)$ pred.
metr_la	2	-0.3	neutral	–
metr_la	3	-0.9	neutral	–
metr_la	6	-5.2	mixing	–
metr_la	12	-3.9	mixing	✓
metr_la	24	-3.5	mixing	✓
metr_la	48	-2.3	mixing	✓
metr_la	96	-4.5	mixing	✓
pems_bay	12	+2.2	phase	×
pems_bay	96	-6.0	mixing	✓
ili_us_state	2	+15.3	phase	✓
ili_us_state	4	+13.7	phase	×
ili_us_state	8	+10.0	phase	×
ili_japan	2	+17.5	phase	×
ili_japan	4	+11.4	phase	×
ili_japan	8	+3.7	phase	×
ili_us_hhs	2	+13.1	phase	×
ili_us_hhs	4	+15.2	phase	×
ili_us_hhs	8	+11.1	phase	×

$\Delta(H)$ sign agrees with outcome in 6/15 settings — no reliable a-priori predictor.

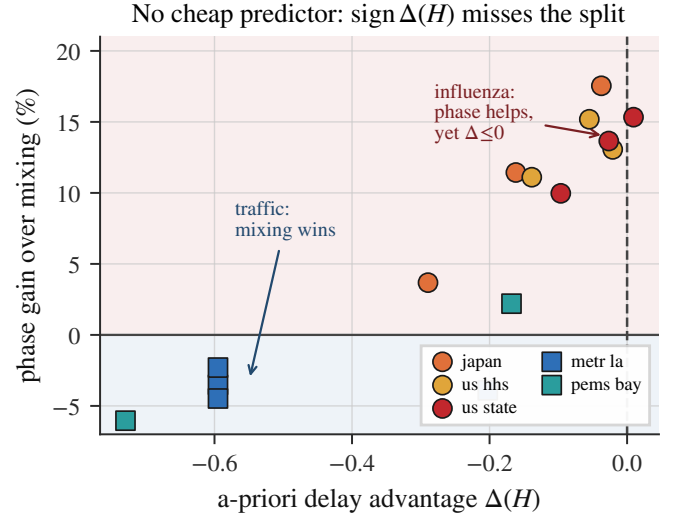


Fig. 3. No cheap predictor: influenza (phase helps, top) sits at $\Delta \approx 0$ while traffic (mixing, bottom) sits at $\Delta \ll 0$; the sign of Δ misses the split.

suite (ETT×4, Weather, Exchange, ILI) is a secondary control on which cross-series structure is weak. The graphs are used only for post-hoc validation, never for training.

b) Protocol: One shared harness: chronological splits; standardisation with train-only statistics; RevIN [49] for neural models; a uniform training budget (30 epochs, early stopping, patience 5) for every model on the traffic and influenza data; MSE loss; Adam. The gate is trained honestly—no post-hoc validation selection of the branch. Look-back $L=96$ (36 for the short weekly series). METR-LA runs the full protocol (3 seeds, all horizons); PEMS-BAY runs a disclosed reduced protocol (1 seed, $H \in \{12, 96\}$) as the confirmation speed benchmark. On the LTSF suite we use the 10-epoch budget standard in that literature so our numbers are comparable to published baselines. We report MSE/MAE on standardised series and Diebold–Mariano tests [50] against the best baseline

TABLE III

Datasets. N : number of series; horizons in native sampling units.

Dataset	Sampling	N	Length	Horizons H	Domain
ETTh1/h2	1 h	7	17 420	{96, 192, 336, 720}	electricity temp.
ETTm1/m2	15 min	7	69 680	{96, 192, 336, 720}	electricity temp.
Weather	10 min	21	52 696	{96, 192, 336, 720}	meteorology
Exchange	1 day	8	7 588	{96, 192, 336, 720}	FX rates
ILI	1 week	7	966	{24, 36, 48, 60}	influenza (US)
METR-LA	5 min	207	34 272	{12, 24, 48, 96}	traffic speed
PEMS-BAY	5 min	325	52 116	{12, 96}	traffic speed
ILL-US-state	1 week	49	360	{2, 4, 8}	influenza (states)
ILL-Japan	1 week	47	348	{2, 4, 8}	influenza (pref.)
ILL-US-region	1 week	10	785	{2, 4, 8}	influenza (HHS)

TABLE IV

Model configuration (fixed across datasets; equal training budget).

Model	Configuration
DLinear/NLinear	moving-avg kernel 25 (DLinear); lr 10^{-3}
PatchTST	patch 16 / stride 8, $d_{\text{model}}=64$, 2 layers, 4 heads; lr 3×10^{-4}
iTransformer	$d_{\text{model}}=128$, 3 layers, 8 heads, FFN 256; lr 3×10^{-4}
VAR	lag order by BIC (≤ 8); low- N datasets only
CSLL	4 bands, $\tau_{\text{max}}=L/2$, rank-16 coupling if $N>64$, dynamic delays, xcorr init, warm gate 0.05; lr 10^{-3}
CSLL2H	CSLL + gated real-mixing branch (Eq. 2)

per setting. All runs are CPU-only for reproducibility.

c) Baselines and configuration: We compare against SeasonalNaive, VAR (where tractable), NLinear, DLinear, PatchTST and iTransformer, all in the shared harness, plus four internal variants that isolate the mechanism: CI-backbone (branch off), phase-off (real mixing, $D \equiv 0$), static (fixed delays), and the full dynamic model. Table IV lists the per-model configuration; CSLL uses 4 frequency bands, $\tau_{\text{max}}=L/2$, and a rank-16 coupling on the high-dimensional datasets.

VII. Results

a) Phase-essential regime: regional influenza: On all three influenza datasets the phase branch robustly improves the lightweight (linear-backbone) model over its real-mixing counterpart (US-state +15%, 0.354 $\rightarrow$ 0.300 at $H=2$; US-region +13%; Japan +17%; three seeds, gap $\gg$ seed noise; Table V), closing most of the distance from the linear baselines to the cross-variate Transformers at $\sim 14\times$ fewer parameters. We do not claim SOTA: iTransformer and PatchTST win outright here (US-state $H=2$: iTransformer 0.252, PatchTST 0.265, CSLL 0.300), because a strong attention backbone already captures the cross-series structure the phase branch supplies to a weak one (Sec. VIII).

b) Sub-horizon regime: traffic: On 5-minute traffic, propagation is sub-horizon, so real-valued mixing captures the dominant same-time congestion structure and the phase adds little. On METR-LA the cheap spectral branch is competitive: it matches or beats DLinear, PatchTST

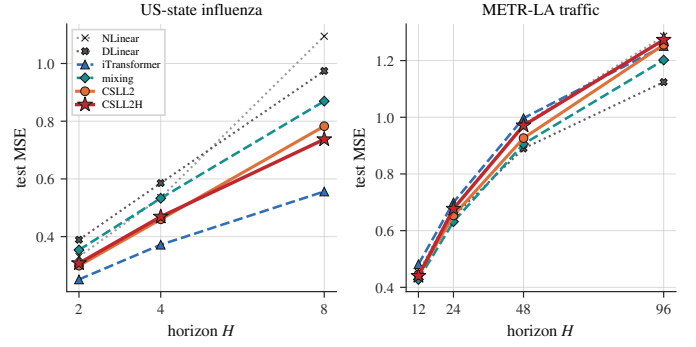


Fig. 4. MSE versus horizon. Left: US-state influenza (phase-essential)—the phase branch separates from real mixing at every horizon. Right: METR-LA traffic (mixing)—phase and mixing coincide; cheap spectral models sit below the Transformers.

and iTransformer at a fraction of their parameters (Table VI, Fig. 5). On the denser PEMS-BAY the cross-variate iTransformer is strongest and CSLL trails it (significantly at $H=96$ by a Diebold–Mariano test); what does carry over is the phase-versus-mixing pattern—the phase branch does not improve on real mixing there either—so PEMS-BAY confirms the same regime, not a CSLL accuracy win.

c) One model across the dichotomy: the hybrid: Pure direct-phase, unable to express same-time coupling, regresses where mixing is what helps—most sharply on the standard benchmarks (Weather/96: phase-only 0.184 vs. mixing 0.156, a 17% regression) and mildly on traffic. The gated hybrid removes this: on Weather/96 it recovers 0.160 (matching the mixing specialist), on traffic it matches the mixing branch within $\sim 1\%$, and on the phase-essential influenza it keeps the phase gain (Table VII). One model, without a regime label, is within seed noise of the best specialised variant everywhere.

d) Horizon behaviour: Fig. 4 plots MSE against horizon in both regimes. On influenza the full model (and its hybrid) tracks the strong baselines while its real-mixing variant lags at every horizon; on traffic the ordering collapses—mixing and phase coincide and all cheap models cluster below the Transformers.

e) Ablation: isolating the components: Table VIII adds the components one at a time over a channel-independent backbone. On influenza, adding real mixing gives little, but adding the phase (static or dynamic) delivers the branch gain; on traffic and Weather the mixing already captures the coupling and the phase adds nothing. The dynamic (input-adaptive) delay is at best marginal over the static one—we report it as neutral rather than a contribution.

f) LTSF control: On the standard benchmarks (Table IX), where cross-series structure is weak, CSLL behaves as a competent CI model: within 5% of the best baseline in half the settings and never far off, i.e. correctly near-inert. The one visible failure is exactly the phase-hurts-mixing effect—on Weather the phase model (0.184 at $H=96$) trails its mixing variant (0.156), and the hybrid

TABLE V

Regional influenza, test MSE (lower better). CSLL beats the linear baselines and its own real-mixing variant; the strong Transformers remain best in absolute terms.

Dataset	H	SNaive	VAR	NLinear	DLinear	PatchTST	iTrans.	CI-only	mixing	CSLL2 _s	CSLL2	CSLL2H
ili_japan	2	0.803	2.057	0.679	0.751	0.588	0.637	0.925	0.852	0.825	0.703	0.735
	4	1.729	4.533	1.443	1.070	1.230	1.325	1.485	1.453	1.492	1.287	1.314
	8	3.182	5.785	2.397	1.362	2.113	2.028	2.243	2.218	2.232	2.136	2.140
ili_us_hhs	2	0.156	0.148	0.145	0.157	0.117	0.138	0.157	0.146	0.143	0.127	0.130
	4	0.304	0.285	0.284	0.282	0.201	0.223	0.289	0.262	0.274	0.223	0.233
	8	0.618	0.549	0.529	0.502	0.301	0.326	0.515	0.459	0.499	0.408	0.405
ili_us_state	2	0.310	1.018	0.331	0.389	0.265	0.252	0.359	0.354	0.362	0.300	0.308
	4	0.560	1.340	0.535	0.586	0.414	0.372	0.538	0.533	0.574	0.460	0.469
	8	1.124	1.753	1.094	0.974	0.620	0.556	0.877	0.869	0.894	0.783	0.737

TABLE VI

Spatio-temporal traffic, test MSE. On METR-LA the cheap spectral model is competitive with far larger baselines; on the denser PEMS-BAY (reduced protocol) iTransformer is strongest, while the phase-versus-mixing pattern is unchanged.

Dataset	H	SNaive	NLinear	DLinear	PatchTST	iTrans.	CI-only	mixing	CSLL2 _s	CSLL2	CSLL2H
metr_la	12	0.498	0.453	0.448	0.445	0.481	0.453	0.428	0.437	0.444	0.440
	24	0.774	0.680	0.650	—	0.700	0.679	0.632	0.650	0.654	0.677
	48	1.163	0.975	0.890	—	0.997	0.969	0.906	0.936	0.926	0.971
	96	1.592	1.286	1.124	1.200	1.251	1.275	1.202	1.233	1.256	1.273
pems_bay	12	0.428	0.405	0.380	—	0.273	0.393	0.332	0.341	0.325	0.320
	96	1.796	1.398	1.015	—	0.690	1.245	0.966	—	1.024	1.243

TABLE VII

The hybrid matches the best specialised branch in each regime with one model (bold = within 2% of the best specialist).

Dataset	H	CI-only	mixing	phase	hybrid
ili_us_state	2	0.359	0.354	0.300	0.308
ili_us_state	8	0.877	0.869	0.783	0.737
ili_japan	2	0.925	0.852	0.703	0.735
metr_la	12	0.453	0.428	0.444	0.440
metr_la	24	0.679	0.632	0.654	0.677
weather	96	—	0.156	0.184	0.162
weather	336	—	0.263	0.285	0.271

TABLE VIII

Ablation (mean test MSE over horizons): components added over the CI backbone. The branch gain vs CI is the full model's improvement over the backbone.

Dataset	CI only (backbone)	+mixing (phase-off)	+static ϕ	+dynamic ϕ (full)	branch gain vs CI (%)
ili_japan	1.551	1.508	1.516	1.375	+14.0
ili_us_hhs	0.320	0.289	0.305	0.253	+21.1
ili_us_state	0.592	0.585	0.610	0.514	+14.0
metr_la	0.844	0.792	0.814	0.820	+2.9
pems_bay	0.819	0.649	0.341	0.675	+17.5

(0.162) recovers most of the gap—which is why the LTSF suite motivates, rather than contradicts, the hybrid.

g) Efficiency and significance: CSLL uses tens of thousands of parameters against hundreds of thousands for the Transformers and trains several times faster (Table X, Fig. 5). Diebold–Mariano tests against the best baseline per setting (Table XI) accompany the point estimates.

h) Interpretability: exact on controlled, partial on real: On planted-delay data the operator recovers delays exactly (Sec. IV). The real datasets carry the structure: a direct cross-correlation analysis of METR-

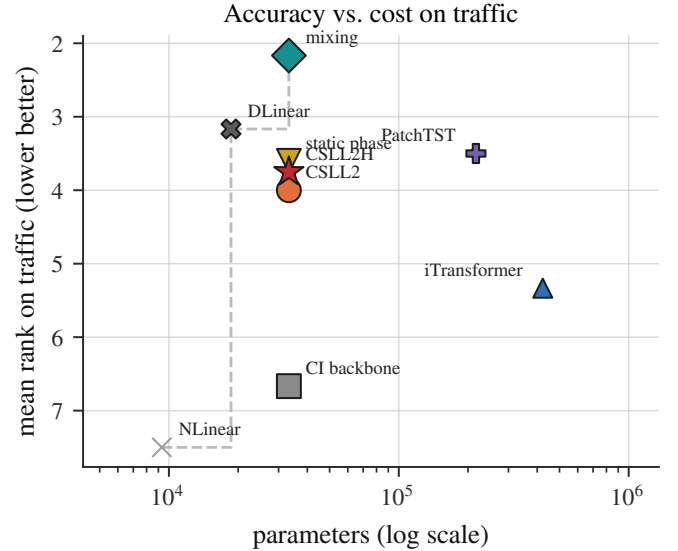


Fig. 5. Accuracy (average rank on traffic) versus parameters. CSLL variants (*) sit at the low-parameter end of the frontier.

LA recovers congestion back-propagation at ≈ 16 km/h, inside the textbook 15–20 km/h kinematic-wave range. The model's learned delays, however, align with observed geography only weakly (Fig. 6): on METR-LA the learned-delay/road-distance relationship is noisy, and on influenza the input-conditioned delays sit at the epidemiologically plausible multi-week scale but concentrate on the border-adjacency graph only slightly. We read this as a limit of the ground truth (road distance and shared borders are imperfect proxies for the effective coupling) as much as the model, and claim plausible but not clean delays on real data.

TABLE IX

Standard LTSF benchmarks, test MSE (10-epoch protocol, comparable to the literature). CSLL is a competent channel-independent model here; the hybrid removes the phase regression (e.g. Weather).

Dataset	H	NLinear	DLinear	PatchTST	iTrans.	mixing	CSLL2	CSLL2H
ETTh1	96	0.397	0.393	0.386	0.392	0.395	0.388	0.393
	192	0.442	0.443	0.437	0.437	0.442	0.438	—
	336	0.484	0.488	0.489	0.480	0.483	0.482	0.478
	720	0.475	0.517	0.581	0.489	0.487	0.483	—
ETTh2	96	0.293	0.311	0.296	0.306	0.343	0.296	—
	192	0.378	0.419	0.375	0.380	0.419	0.381	—
	336	0.421	0.533	0.418	0.422	0.453	0.429	—
	720	0.429	0.706	0.451	0.434	0.450	0.446	—
ETTm1	96	0.354	0.345	0.322	0.341	0.344	0.344	—
	192	0.391	0.383	0.357	0.389	0.384	0.385	—
	336	0.423	0.415	0.385	0.439	0.414	0.416	—
	720	0.487	0.478	0.448	0.507	0.480	0.479	—
ETTm2	96	0.182	0.193	0.176	0.186	0.178	0.181	—
	192	0.247	0.263	0.243	0.252	0.243	0.247	—
	336	0.308	0.358	0.306	0.315	0.305	0.308	—
	720	0.407	0.474	0.408	0.420	0.408	0.408	—
exchange	96	0.084	0.082	0.088	0.092	0.087	0.083	0.083
	192	0.176	0.154	0.190	0.184	0.180	0.177	—
	336	0.328	0.280	0.349	0.335	0.349	0.343	0.339
	720	0.912	0.677	0.901	0.856	0.990	0.903	—
ili	24	2.827	3.056	2.298	2.352	2.598	2.520	—
	36	2.675	2.936	2.340	2.075	2.465	2.429	—
	48	2.562	2.938	2.253	2.279	2.443	2.415	—
	60	2.584	3.068	2.160	2.133	2.491	2.468	—
weather	96	0.193	0.194	0.163	0.166	0.156	0.184	0.162
	192	0.241	0.239	0.208	0.213	0.206	0.232	0.216
	336	0.292	0.285	0.265	0.275	0.263	0.285	0.271
	720	0.366	0.350	0.345	0.352	0.342	0.358	0.349

TABLE X

Parameter count and mean training time (tractable datasets).

Model	Params	Train time (s)
SNaive	—	0.0
VAR	—	0.6
NLinear	9.3k	4.7
DLinear	18.6k	9.7
PatchTST	216.5k	306.8
iTrans.	422.9k	1333.0
CSLL2	33.3k	115.3

VIII. Discussion and Limitations

The dichotomy gives a design rule of thumb: phase/delay structure helps on data with delayed cross-series coupling (epidemics, long-range propagation) and not on same-time-dominated data; since simple statistics do not reliably tell the two apart in advance, the gated hybrid applies the rule automatically by learning the blend. A second finding sets the scope of the mechanism’s value: a strong cross-variate backbone subsumes it. The phase

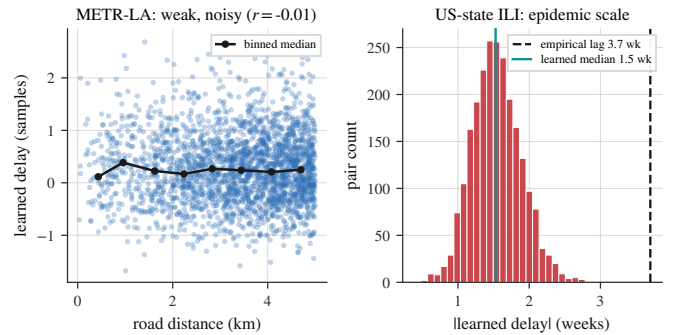


Fig. 6. Learned delays on real data. Left: METR-LA learned delay vs. road distance (noisy). Right: US-state ILI input-conditioned delay magnitudes at the epidemiologically plausible scale (dashed: empirical inter-region lag).

branch improves a lightweight CI backbone by 13–19% on epidemic data, but grafting it onto an iTransformer backbone yields no gain—the attention layers already capture the delayed structure the explicit operator supplies to a weak backbone. Phase-as-delay is thus most valuable

TABLE XI
Diebold–Mariano tests, CSLL vs. best baseline per setting
($\succ/\approx/\prec$: CSLL significantly better / tied / worse).

Dataset	H	best base	CSLL2	base	sig.
metr_la	12	DLinear	0.444	0.444	$\succ$
metr_la	24	DLinear	0.647	0.653	$\succ$
metr_la	48	DLinear	0.926	0.891	$\succ$
metr_la	96	DLinear	1.253	1.138	$\succ$
pems_bay	12	iTrans.	0.324	0.273	$\approx$
pems_bay	96	iTrans.	1.024	0.690	$\succ$
ili_us_state	2	iTrans.	0.295	0.240	$\approx$
ili_us_state	4	iTrans.	0.485	0.368	$\succ$
ili_us_state	8	iTrans.	0.759	0.522	$\succ$
ili_japan	2	PatchTST	0.680	0.588	$\approx$
ili_japan	4	DLinear	1.312	1.091	$\succ$
ili_japan	8	DLinear	2.133	1.383	$\succ$
ili_us_hhs	2	PatchTST	0.127	0.117	$\approx$
ili_us_hhs	4	PatchTST	0.223	0.201	$\succ$
ili_us_hhs	8	PatchTST	0.416	0.301	$\succ$

where capacity is scarce: a cheap way to give a small model the cross-series reach of a large one.

Limitations, stated plainly. (i) CSLL does not beat the strongest baselines in absolute accuracy; its case is efficiency, adaptivity, and a characterised mechanism. (ii) We do not provide an a-priori predictor of the regime: the lead statistics we tried fail (Sec. V); a reliable cheap predictor remains open. (iii) Interpretability transfers imperfectly to real data (Sec. VII). (iv) The dynamic delay variant helps little; we report it as neutral. (v) The rFFT assumes within-window quasi-stationarity and bounds delays by $\tau_{\max}$. (vi) PEMS-BAY uses a disclosed reduced protocol (1 seed, $H \in \{12, 96\}$); we do not claim per-setting significance there, only the qualitative regime. We do not benchmark against pretrained foundation models, whose corpora may overlap our test data.

a) Future work: Three directions follow directly. First, a reliable a-priori regime predictor: our lead statistics fail, but a small validation probe (train both branches briefly and read the gates) may substitute cheaply for a closed-form statistic. Second, multi-modal delays: the current operator favours a dominant lead per pair, whereas epidemics and transport networks carry a distribution of lags; a mixture of delay components per band is a natural extension. Third, delay-aware pretraining: because the operator is cheap and interpretable, it could serve as a structured, low-parameter cross-series head on top of a frozen univariate foundation model, combining the reach of pretraining with an explicit, auditable coupling.

b) Broader impact: The datasets we study—traffic and epidemics—are decision-relevant, and an interpretable delay matrix is attractive for operators who must justify forecasts. We caution against the converse: the learned delays are predictive alignments, not causal claims, and Sec. VII shows they align only weakly with physical geography on real data. Reading them as causal transmission paths would be a misuse.

IX. Conclusion

We turned a physical prior—that inter-series coupling carries frequency-dependent propagation delays—into an exact, identifiable, cheap operator, and used it to answer when phase-aware cross-series modelling pays. The mechanism recovers planted delays exactly, helps a lightweight model on genuinely lagged data (regional epidemics), and is redundant on same-time data; because no cheap statistic predicts which regime holds, a gated hybrid learns the blend and is safe and near-best across regimes with one model. We report where the mechanism helps, where it is redundant, and where a strong backbone subsumes it, rather than a single headline number.

Reproducibility

All datasets are public with fetch scripts and checksums; code, configs, fixed seeds, and scripts to regenerate every table and figure accompany the paper.

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